

Python

File Edit Selection View Go Run Terminal Help

Customer_Analysis.py

```
17 print(df[['age', 'age_group']].head(5)) df = customer_id age gender ... payment_method frequency_of_purchases
18 print(df['frequency_of_purchases'].unique())
19 frequency = {
20     'Annually': 365,
21     'Quarterly': 90,
22     'Weekly': 7,
23     'Monthly': 30,
24     'Bi-weekly': 3.5,
25     'Fortnightly': 14,
26     'Every 3 Months': 90
27 }
28 df['frequency_in_days'] = df['frequency_of_purchases'].map(frequency)
29 print(df[['frequency_of_purchases', 'frequency_in_days']].head(5))
30
```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS

```
'payment_method', 'frequency_of_purchases'],
dtype='str')
age age_group
0 55 Middle-aged
1 19 Young Adults
2 50 Middle-aged
3 21 Young Adults
4 45 Middle-aged
<StringArray>
[ 'Fortnightly', 'Weekly', 'Annually', 'Quarterly',
'Bi-weekly', 'Monthly', 'Every 3 Months']
Length: 7, dtype: str
frequency_of_purchases frequency_in_days
0 Fortnightly 14.0
1 Fortnightly 14.0
2 Weekly 7.0
3 Weekly 7.0
4 Annually 365.0
```

Python Debugger: Python File (Python)

Ln 29, Col 1 Spaces: 4 UTF-8 CRLF Python Python 3.13

```
1 import cProfile
2 import pydoc
3 import pandas as pd
4 df = pd.read_csv('C:\\Users\\josep\\Downloads\\Python\\customer_shopping_behavior.csv')
5 print(df.head())
6 df.info()
7 print(df.describe(include='all'))
8 print(df.isnull().sum())
9 df['Review Rating'] = df.groupby('Category')['Review Rating'].transform(lambda x: x.fillna(x.median()))
10 print(df.isnull().sum())
11 df.columns = df.columns.str.lower()
12 df.columns = df.columns.str.replace(' ', '_')
13 df.columns = df.columns.str.replace('purchase_amount_(usd)', 'purchase_amount')
14 print(df.columns)
15 label = ['Young Adults', 'Adults', 'Middle-aged', 'Seniors']
16 df['age_group'] = pd.qcut(df['age'], q=4, labels=label) label = ['Young Adults', 'Adults', 'Middle-aged', 'Seniors']
17 print(df[['age', 'age_group']].head(5)) df = customer_id age gender ... payment_method frequency_of_purchases
18
```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS

```
Discount Applied      0
Promo Code Used       0
Previous Purchases    0
Payment Method        0
Frequency of Purchases 0
dtype: int64
Index(['customer_id', 'age', 'gender', 'item_purchased', 'category',
      'purchase_amount', 'location', 'size', 'color', 'season',
      'review_rating', 'subscription_status', 'shipping_type',
      'discount_applied', 'promo_code_used', 'previous_purchases',
      'payment_method', 'frequency_of_purchases'],
      dtype='str')
age age_group
0 55 Middle-aged
1 19 Young Adults
2 50 Middle-aged
3 21 Young Adults
4 45 Middle-aged
```

Python Debugger: Python File (Python)

```

17 print(df[['age', 'age_group']].head(5)) df = customer_id age gender ... payment_met
18 print(df['frequency_of_purchases'].unique())
19 frequency = {
20     'Annually': 365,
21     'Quarterly': 90,
22     'Weekly': 7,
23     'Monthly': 30,
24     'Bi-weekly': 3.5,
25     'Fortnightly': 14,
26     'Every 3 Months': 90}
27 df['frequency_in_days'] = df['frequency_of_purchases'].map(frequency)
28 print(df[['frequency_of_purchases', 'frequency_in_days']].head(5))
29
30

```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS

```

    'payment_method', 'frequency_of_purchases'],
    dtype='str')
  age    age_group
0    55  Middle-aged
1    19  Young Adults
2    50  Middle-aged
3    21  Young Adults
4    45  Middle-aged
<StringArray>
[ 'Fortnightly',      'Weekly',      'Annually',      'Quarterly',
  'Bi-Weekly',      'Monthly', 'Every 3 Months']
length: 7, dtype: str
  frequency_of_purchases  frequency_in_days
0          Fortnightly           14.0
1          Fortnightly           14.0
2              Weekly              7.0
3              Weekly              7.0
4          Annually           365.0

```

The screenshot shows the VS Code Python IDE with a Jupyter Notebook. The script in the notebook is as follows:

```

1 import cProfile
2 import pydoc
3 import pandas as pd
4 df = pd.read_csv('c:\Users\joseph\Downloads\Python\customer_shopping_behavior.csv')
5 print(df.head())
6 df.info()
7 print(df.describe(include='all'))
8 print(df.isnull().sum())
9 df['Review Rating'] = df.groupby('Category')['Review Rating'].transform(lambda x: x.fillna(x.median()))
10 print(df.isnull().sum())
11 df.columns = df.columns.str.lower()
12 df.columns = df.columns.str.replace(' ', '_')
13 df.columns = df.columns.str.replace('purchase_amount_usd', 'purchase_amount')
14 print(df.columns)
15 label = ['Young Adults', 'Adults', 'Middle-aged', 'Seniors']
16 df['age_group'] = pd.cut(df['age'], q=4, labels=label)
17 print(df[['age', 'age_group']].head(5))
18

```

The output of the script shows the first 5 rows of the data:

	customer_id	age	gender	item_purchased	category	purchase_amount	location	size	color	season	review_rating	subscription_status	shipping_type	discount_applied	promo_code_used	previous_purchases	payment_method	frequency_of_purchases
0	55	Middle-aged	1	19	Young Adults	2	50	Middle-aged	3	21	Young Adults	4	45	Middle-aged				